Milestone 1

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Topic

Short-Term Air Quality Forecasting Using Time-Series and Machine Learning Models

Dataset(s)

https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data (10+ datasets 2013-2017 every day/hour recordings of air quality)

Framework

This project follows a machine learning forecasting framework. The workflow induces preprocessing of the hourly air quality and meteorological data which will be used to predict short-term air quality, such as PM2.5 or PM10 for the next 1 hour, 3 hours, 6 hours, 12 hours and 24 hours. We will compare models using metrics such as MAE or RMSE and we will analyze which variables contribute most to the forecasting accuracy. Furthermore, the team is proposing to split the work into 5 phases:

1. ML Data Pipeline: Handle missing values using imputation (such as KNN). Cyclical features will also need to be encoded. And scaling methods need to be applied to features such as PM2.5.
2. Temporal feature engineering: The team would have to construct historical lagged input vectors which would translate static rows into a sequential format. We need this, because the model will require memory (if conditions from 5PM are fed, it must also be aware of what happened at 4PM). The team might also explore the usage of rolling statistics (create windows of the 12 hour average for example)
3. ML Architectures: Train and adjust hyperparameters to dual approaches: (XGBoost, LightGBM, etc.) to capture non-linear input space interactions and Deep Learning models such as LSTMs to find long-range over time dependencies.
4. Time-Aware evaluation: instead of using regular K-Fold cross validation, the team proposes to use walk-forward cross validation to prevent data leakage. Predictions would be executed across different levels (1 hour, 3hours, 6, 12, 24 etc.) and would use either Direct or Recursive multi step strategies. The team would then compare accuracy using MAE or RMSE as stated before.
5. Explainable AI: Lastly, we propose to move the models ‘’black box’’ to a transparent system by utilizing the feature importance scores. The team can then use these scores to rank which variables are best used for making forecasts.

Supportive Literature:

Abuouelezz, W., Ali, N., Aung, Z., Altunaiji, A., Shah, S. B., & Gliddon, D. (2025). Exploring PM2.5 and PM10 ML forecasting models: a comparative study in the UAE. Scientific Reports, 15(1). https://doi.org/10.1038/s41598-025-94013-1

Peinado, L. B., Guarda, T., Germán Herrera-Vidal, Minnaard, C., & Coronado-Hernández, J. R. (2025). Statistical and Machine Learning Models for Air Quality: A Systematic Review of Methods and Challenges. Algorithms, 18(12), 783–783. https://doi.org/10.3390/a18120783

Méndez, M., Merayo, M. G., & Núñez, M. (2023). Machine learning algorithms to forecast air quality: a survey. Artificial Intelligence Review. https://doi.org/10.1007/s10462-023-10424-4